SOFTWARE TEST PLAN

Vehicle Parking System

Item	Details
Project	Vehicle Parking System (VPS)
Team	10
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Version	1.0
Date	16-09-2026
Status	Draft

1. Introduction

Purpose: This document defines the test plan for the Vehicle Parking System version 1.0. It describes what has to be tested, the testing approach, test environment, schedule, responsibilities and expected test deliverables.

Scope: Testing covers user registration and login, parking slot viewing, booking, vehicle details, check-in and check-out, booking history, and administrator functions. Payment processing, parking sensors, cameras and physical gate hardware are not part of this version.

References: Vehicle Parking System SRS v1.0 and the Software Test Plan template provided for the course.

Definitions: VPS - Vehicle Parking System; SRS - Software Requirements Specification; STP - Software Test Plan; RTM - Requirements Traceability Matrix; UAT - User Acceptance Testing.

2. Test Items

- User registration and login
- Parking slot display and floor grouping
- Parking slot booking
- Vehicle registration during booking
- Vehicle check-out
- Booking history
- Admin parking slot management
- Admin active booking and force checkout

- Session and access control

3. Features to be Tested

The following features are taken from the Vehicle Parking System SRS and are covered in the test plan.

Requirement ID	Feature	Test Case
VPS-F-001	User registration	TC-Reg-01
VPS-F-002	Duplicate email validation	TC-Reg-02
VPS-F-003	Password hashing	TC-Auth-01
VPS-F-004	User authentication	TC-Auth-02
VPS-F-005	Display parking slots	TC-Slot-01
VPS-F-006	Floor-wise slot display	TC-Slot-02
VPS-F-007	Prevent booking occupied slot	TC-Slot-03
VPS-F-008	Book available slot	TC-Book-01
VPS-F-009	Register new vehicle	TC-Book-02
VPS-F-010	Use existing vehicle record	TC-Book-03
VPS-F-011	Record check-in time	TC-Book-04
VPS-F-012	Customer check-out	TC-Checkout-01
VPS-F-013	Check-out authorization	TC-Checkout-02
VPS-F-014	View booking history	TC-Hist-01
VPS-F-015	Admin add parking slot	TC-Admin-01
VPS-F-016	Admin remove available slot	TC-Admin-02
VPS-F-017	Admin view/force checkout	TC-Admin-03

Non-functional requirements:

Requirement ID	Requirement	Test / Review
VPS-NF-001	Parking dashboard response time <= 2 seconds for up to 200 slots	TC-Perf-01
VPS-NF-002	Core actions should be reachable within 3 clicks	TC-UX-01
VPS-NF-003	Slot status should remain consistent with active bookings	TC-Rel-01
VPS-NF-004	Session should expire after logout	TC-Sec-01
VPS-NF-005	Models, routes and templates should be separated	CR-01

Security requirements:

Requirement ID	Security Feature	Test Case
VPS-SR-001	Secure password hashing	TC-Sec-02
VPS-SR-002	Admin route authorization	TC-Sec-03

VPS-SR-003	Session invalidation	TC-Sec-01
VPS-SR-004	Input validation and sanitisation	TC-Sec-04
VPS-SR-005	Checkout allowed only for owner/admin	TC-Checkout-02

4. Features Not to be Tested

- Payment gateway or online payment processing.
- Parking sensors, cameras or automatic vehicle detection.
- Physical parking gate or barrier hardware.
- External banking systems.
- Production infrastructure not included in the project.

5. Test Approach / Strategy

Levels:

- Unit testing - testing individual functions and modules.
- Integration testing - checking the interaction between Flask routes, database and session handling.
- System testing - testing complete customer and admin workflows.
- Acceptance testing (UAT) - checking the system against the requirements and acceptance criteria.

Types:

- Functional testing
- Regression testing
- Performance testing
- Usability testing
- Reliability testing
- Security testing
- Database/data integrity testing

Entry Criteria: Stable build delivered, test data available, application and database ready, and test environment available.

Exit Criteria: All planned test cases executed, all high-priority requirements tested, no critical defects open for submission, and acceptance criteria satisfied.

5.1 Security Validation

- Check that passwords are not stored in plain text.
- Check that non-admin users cannot access admin functions.
- Check that logout invalidates the user session.
- Check that invalid or unsafe form input is rejected or handled safely.
- Check that one customer cannot check out another customer's booking.
- HTTPS will be checked for production deployment.

6. Test Environment

Item	Details
Hardware	Normal laptop/desktop or mobile device
Operating System	Any OS supporting Python 3
Application	Python 3, Flask
Database	SQLite using Flask-SQLAlchemy
Browser	Chrome / Edge / Firefox
Test Data	Dummy users, vehicles, parking slots and bookings
Performance Data	Up to 200 parking slots
Defect Tracking	Team defect log / Jira if used

7. Test Schedule

Activity	Date	Output
Test case preparation	15-Sep-2026	Test cases and requirement mapping
Environment setup	16-Sep-2026	Application and database ready
Unit and integration testing	17-18 Sep 2026	Module test results
System testing	19-20 Sep 2026	End-to-end test results
Security, performance and usability testing	21-Sep-2026	NFR/security results
Defect retesting	22-Sep-2026	Retest results
Final/UAT testing	23-Sep-2026	Final acceptance results

8. Test Deliverables

- Software Test Plan
- Test cases
- Test data
- Test execution results
- Defect reports
- RTM/coverage report
- Final Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Swapna	Prepare the test plan, coordinate testing and maintain test status.
Test Engineer	Shreya	Prepare and execute test cases and record defects.

Developer	-	Fix reported defects and support testing.
Course Evaluator	-	Review testing documents and final results.

10. Risks and Mitigation

Risk	Mitigation
Unstable build	Run basic smoke tests first and use a stable build for full testing.
Database errors	Keep backup test data and check booking/slot status after each major operation.
Limited time	Test high-priority requirements first.
Environment problems	Keep the setup steps and test database ready for reuse.
Defects found late	Start unit and integration testing early.
Slow application	Test with up to 200 slots and check database/page performance.

11. Assumptions & Dependencies

- The Vehicle Parking System SRS is the baseline for testing.
- The Flask application and SQLite database will be available.
- Dummy test data will be used.
- Admin credentials/role will be available for admin testing.
- No payment or hardware integration is required for this version.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable.
- The build is too unstable to test the main functions.
- A critical defect blocks more than 30% of the planned test cases.
- Test data or the database is corrupted.

Resume testing when:

- Blocking defects are fixed or a workaround is available.
- The test environment is working.
- Test data/database is restored.
- A new build passes smoke testing.

13. Test Case Management & Traceability

The RTM is used to connect every SRS requirement with its corresponding test case. The planned mappings are:

Requirement	Test Case	Status
VPS-F-001 to VPS-F-004	TC-Reg-01, TC-Reg-02, TC-Auth-01, TC-Auth-02	Not Run
VPS-F-005 to VPS-F-007	TC-Slot-01 to TC-Slot-03	Not Run
VPS-F-008 to VPS-F-011	TC-Book-01 to TC-Book-04	Not Run
VPS-F-012 to VPS-F-014	TC-Checkout-01, TC-Checkout-02, TC-Hist-01	Not Run
VPS-F-015 to VPS-F-017	TC-Admin-01 to TC-Admin-03	Not Run
VPS-NF-001 to VPS-NF-005	TC-Perf-01, TC-UX-01, TC-Rel-01, TC-Sec-01, CR-01	Not Run
VPS-SR-001 to VPS-SR-005	TC-Sec-02, TC-Sec-03, TC-Sec-01, TC-Sec-04, TC-Checkout-02	Not Run

14. Test Metrics & Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage passed/failed
- Number of open defects
- Defect severity
- Requirement coverage

Reports:

- Test execution status
- Defect report
- Requirement coverage/RTM
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Swapna	
Developer / Dev Lead	-	
Course Coordinator	Mr Dhiroj Kumar Behera	